Atrijo Pal

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (ECE)	Indian Institute of Technology, Guwahati	8.76 (Current)	2025 – Present
Senior Secondary	CBSE Board	93.4%	2025
Secondary	ICSE Board	98.6%	2023

Experience

- **KensaraAI – B2B Privacy-Compliance SaaS (Azure)** *2025 – Present*
  AI Engineer (Full-Stack) Remote
  - Sole engineer on the **M1 Data Intelligence Core**: shipped **7 production modules** (RoPA Engine, Schema Builder, Document Vault, Access Control, Communication Centre) as a **multi-tenant** B2B SaaS on Azure
  - Designed the backend in **FastAPI + asyncpg**: a **30+ table PostgreSQL schema** with row-level tenant isolation, immutable audit logs and an async task architecture
  - Built a **4-jurisdiction RoPA engine** (GDPR/DPDPA/CCPA/PDPL) with Kimi K2 LLM generation, a **RAG pipeline** over regulatory texts (Qdrant + Voyage AI) and a **Text-to-SQL AI assistant**

Projects

- **SEM Image Restoration: Denoising + 2× Super-Resolution** *2026*
  KLA AI Hackathon 2026 | Team ECE_Lords, IIT Guwahati **Top 5 Finalist** | GitHub
  - Built a joint **denoising + 2× super-resolution** model for noisy, low-res semiconductor SEM images: a **29M-param NAFNet** scoring **23.5 dB PSNR / 0.632 SSIM / 0.143 LPIPS** vs 20.2 dB / 0.500 for bicubic (**+3.3 dB**)
  - Analysed 1,197 image pairs before modelling: AIC-ranked 5 noise models to show the noise is a **Poisson + speckle mixture** (R^2 0.999), confirmed the bicubic degradation and checked for train/test leakage
  - Beat classical baselines (**Wiener, Richardson-Lucy, VST+BM3D**; best 23.08 vs 23.34 dB); gains shrink from **+5.05 to +1.59 dB** on fine textures; traced a checkerboard artifact (**85× spike**) to PixelShuffle
  - Redesigned the model (native-resolution trunk, resize-conv decoder, physics-informed VST, non-local attention): **9.6× faster** (30→290 img/s) for a 1.2 dB cost; fp16 gives NAFNet a lossless **2.5× speedup**; fp4 rejected (**-2.3 dB**)

- **PCOP: Predictive Customer Outreach Platform** *2026*
  Union Bank IDeATHON 2.0 (PSBs Hackathon Series 2026) | Team MoneyLords **2nd Place** | GitHub | Live Demo
  - Built a **7-layer AI platform** (8 microservices) that lets retail banks detect churn early, act on it and **prove the outcome**: existing CRM tools predict who may leave but cannot show which retentions the outreach actually caused
  - **Detection + scoring:** tracks 18 behavioural streams per customer with change-point detection and FDR control (**false alarms 37% → 5%**); a 5-model ensemble (GRU, XGBoost, GraphSAGE, DeepHit) reaches **AUC 0.85–0.93** on public benchmarks, with 90% confidence intervals
  - **Action + proof:** a LangGraph agent (15 tools) picks the next-best-action and drafts outreach for 5 channels, with mandatory human approval; a doubly-robust uplift engine with holdout groups measures true causal retention; DPDPA/TRAI/RBI compliance built in; modelled **11.6× ROI** (INR 128 Cr/yr for a 2.5 Cr-customer bank)

- **BDH-ECHR: Post-Transformer Legal AI** *2026*
  Kriti 2026 Inter-Hostel Hackathon, IIT Guwahati **1st Prize**
  - Trained a Dragon Hatchling (BDH) post-transformer for multi-label ECHR article-violation classification: **Micro-F1 0.7716** at 20.4M params, **+13.2 F1 over BERT-base** and on par with Longformer at **5.4× fewer parameters**
  - Ran a neuron-to-concept interpretability audit: **218 interpretable neurons across 477 legal concepts**, and **4.5× sparser** activations than transformer MLPs

Technical Skills

- **Languages**: Python, C++, C, Java, TypeScript, SQL
- **ML / AI**: PyTorch, XGBoost, CNNs, GNNs (PyTorch Geometric), NLP, LangGraph, RAG, Vector DBs (Qdrant, ChromaDB), LLM Tool-Calling, Causal Inference, Model Optimization (fp16, `torch.compile`)
- **Backend & Systems**: FastAPI, PostgreSQL, Redis, Kafka, Docker, Azure, MLflow, Git, Linux
- **Frontend & Core**: React, TypeScript, Next.js, Tailwind CSS; Data Structures & Algorithms

Positions of Responsibility

- **Deputy Coordinator, ML Module**, Coding Club IIT Guwahati *2025 – Present*
- **Associate**, IITG.ai *2025 – Present*

Achievements

- **Hackathons:** 1st – Kriti 2026 | 2nd – Union Bank IDeATHON 2.0 | Top 5 Finalist – KLA AI Hackathon 2026
- **Academics:** JEE Advanced AIR 1649 | JEE Main 99.701 percentile | WBJEE Rank 26 | JBNSTS State Rank 13
- **Competitive:** Codeforces Rating 1482 | NSEP & NSEC Top 1% (State) | Qualified RMO